

TASK 6 — FINAL PROJECT REPORT

INTRODUCTION

Student Performance Prediction Using Machine Learning

Introduction

Student academic performance is an important indicator of educational success. Academic performance can be influenced by several factors, including study habits, previous academic performance, absences, family background, social activities, and health.

In this project, machine learning is used to analyze student-related data and predict whether a student is likely to pass or fail.

The project follows the complete machine learning workflow, beginning with data collection and preprocessing and continuing through exploratory data analysis, model development, prediction, and evaluation.

Project Goal

The primary goal of this project is to develop a machine learning classification model capable of predicting student performance based on selected features.

PROBLEM STATEMENT & OBJECTIVES

Problem Statement

Educational institutions need effective methods for identifying students who may be at risk of poor academic performance.

This project addresses the problem by developing a machine learning model that predicts student performance based on demographic, academic, behavioral, and other relevant features.

Objectives

- Collect student performance data.
 - Perform data cleaning.
 - Handle categorical variables.
 - Select relevant features.
 - Create a Pass/Fail target.
 - Perform exploratory data analysis.
 - Build a Random Forest classification model.
 - Generate predictions.
 - Evaluate the model.
 - Analyze the results.
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DATASET

Dataset Description

The project uses the Student Performance Dataset.

The dataset contains information related to students' academic, demographic, social, and behavioral characteristics.

Important Features

Feature	Description
school	Student's school
sex	Student's gender
age	Student's age
studytime	Weekly study-time category

failure s	Previous failures
absence s	Number of absences
Medu	Mother's education
Fedu	Father's education
goout	Social activity frequency
health	Health condition
G3	Final grade

Target

```
G3 >= 10 → Pass
```

```
G3 < 10 → Fail
```

```
df.head()
```

DATA PREPROCESSING

Data Cleaning

The following preprocessing steps were performed:

1. Dataset was loaded using Pandas.
2. Dataset dimensions were checked.
3. Data types were examined.
4. Missing values were checked.

5. Duplicate records were checked and removed where necessary.
6. Relevant features were selected.
7. Final grade was converted into a binary target.
8. Categorical variables were converted into numerical form.
9. Data was divided into training and testing sets.

Before Processing

Insert screenshot of:

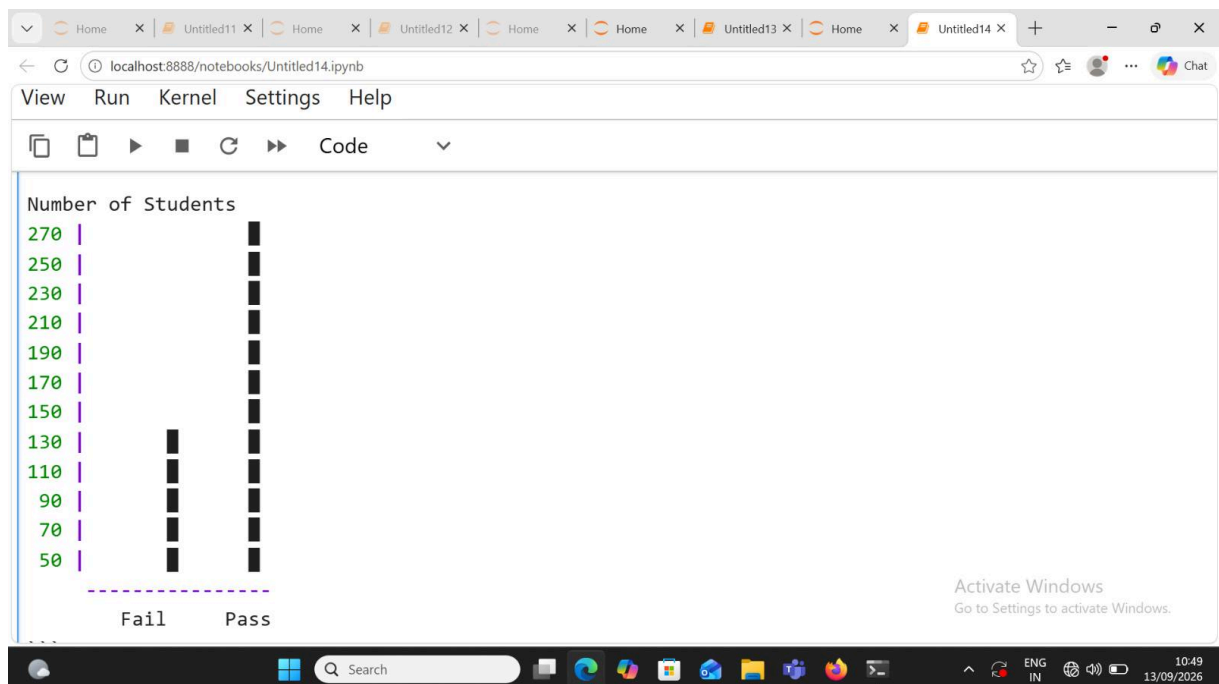
```
df.info()  
df.isnull().sum()  
df.head()
```

```
X.head()
```

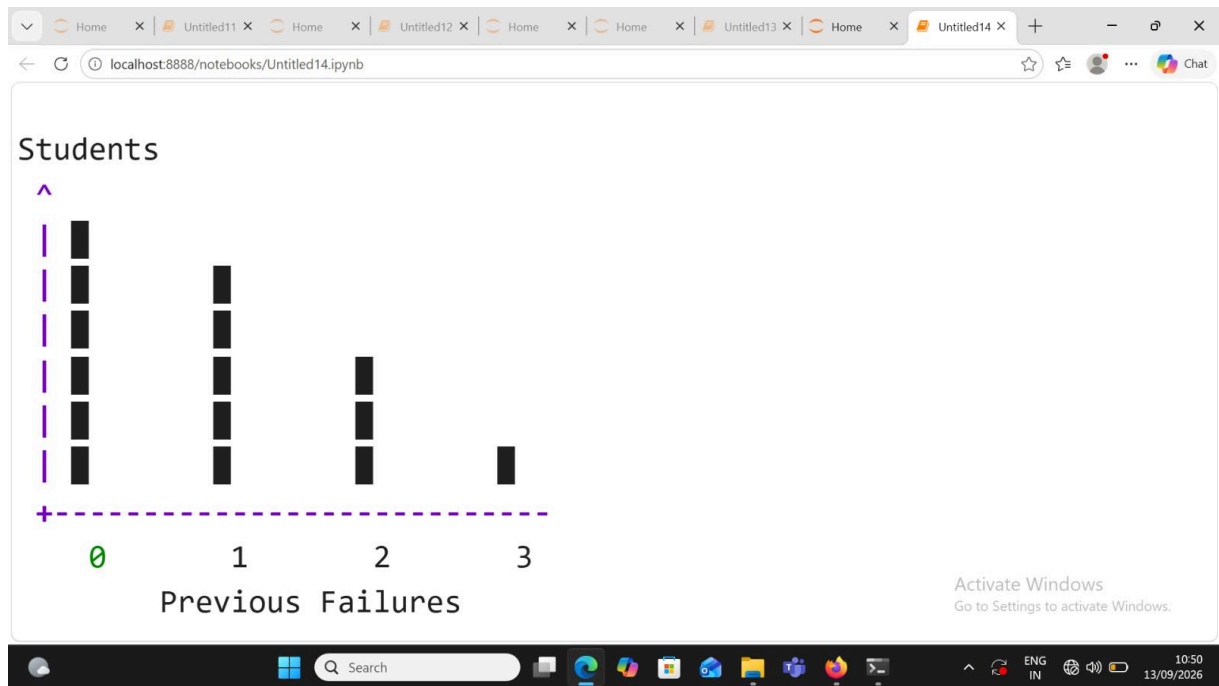
```
print(X.shape)  
print(y.shape)
```

EXPLORATORY DATA ANALYSIS

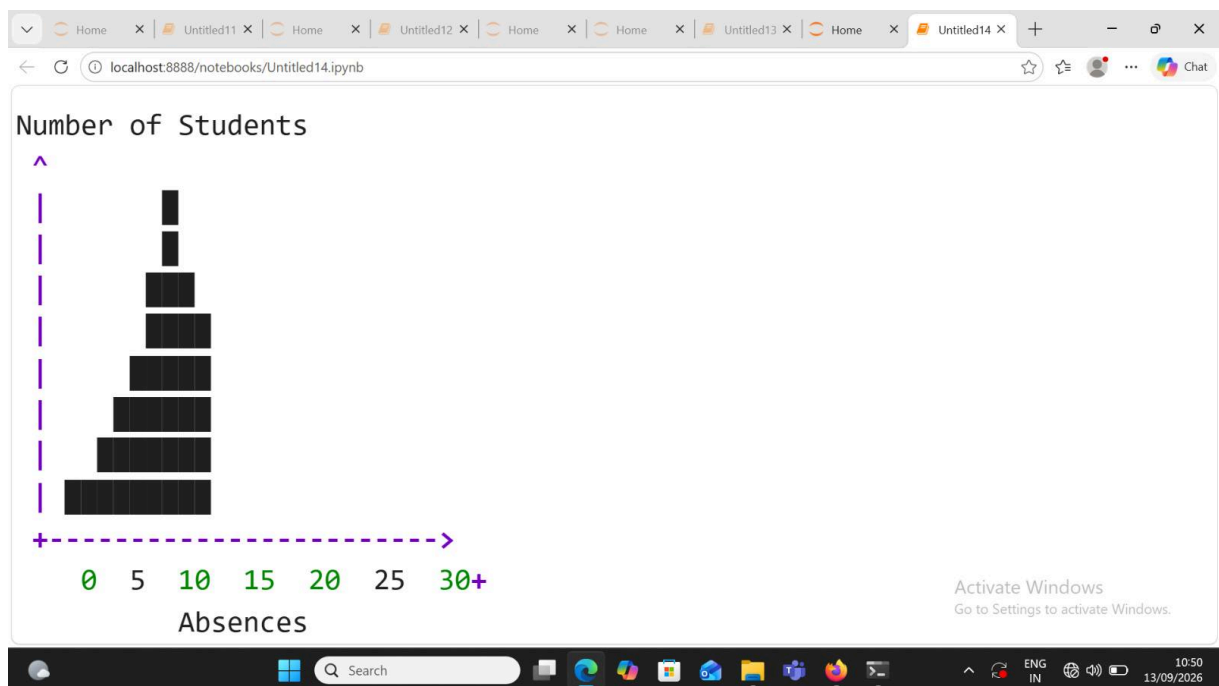
Student Performance Distribution



Study Time vs Performance

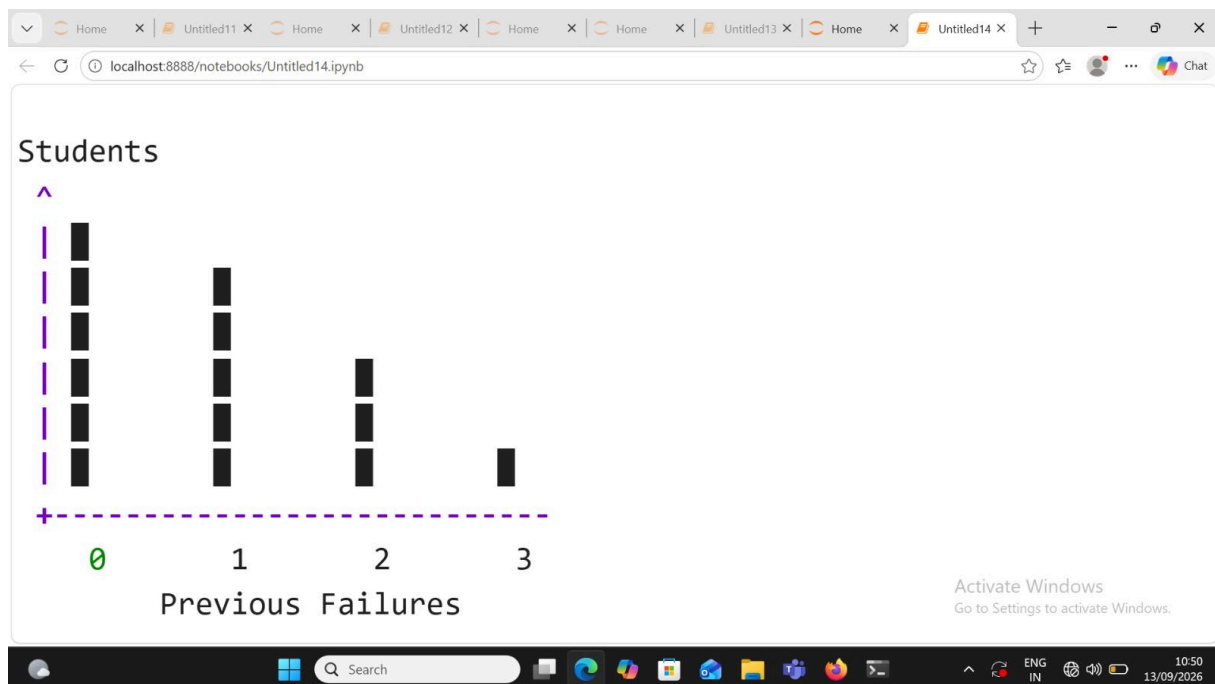


Absence Distribution

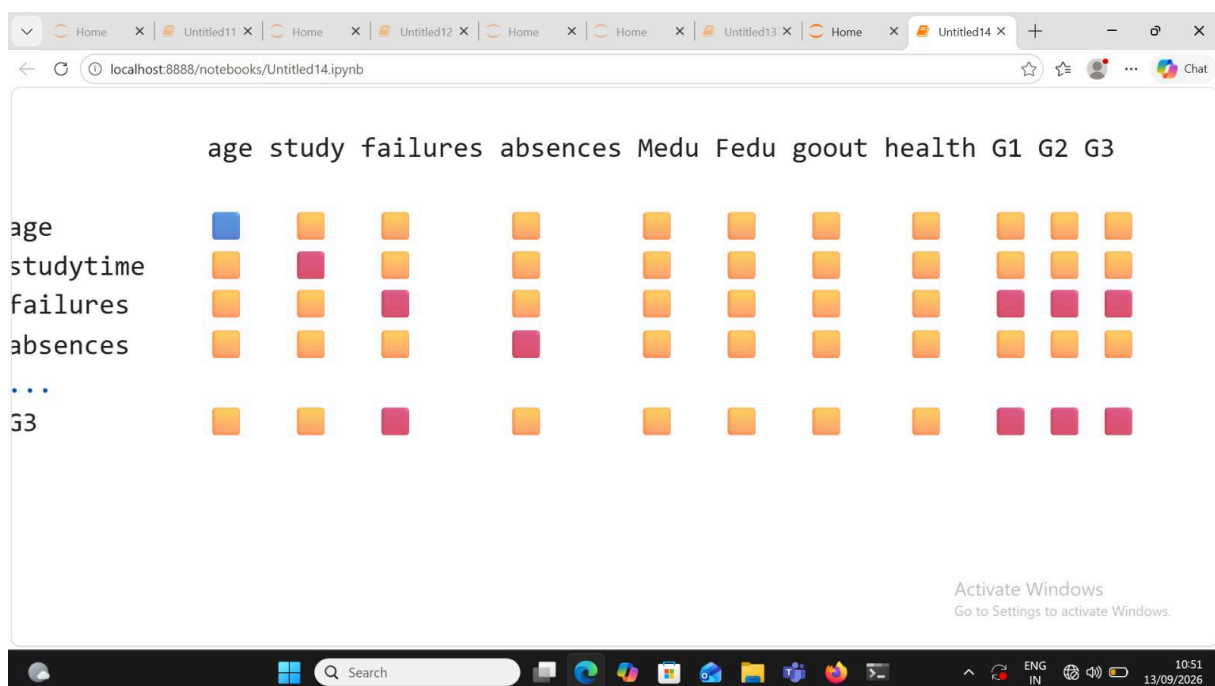


EDA CONTINUED

Previous Failures vs Performance



Correlation Heatmap



Overall EDA Finding

The exploratory analysis provided useful information about the relationship between student characteristics and academic performance. Factors such as study time, previous failures, absences, and other variables were examined to understand their relationship with student outcomes.

MODEL DEVELOPMENT & TRAINING

Selected Algorithm

Random Forest Classifier

Random Forest was selected because it is suitable for classification problems involving tabular data and can model nonlinear relationships between features.

Train-Test Split

The dataset was divided into:

```
80% → Training  
20% → Testing
```

Training

```
model = RandomForestClassifier(  
    n_estimators=100,  
    random_state=42  
)  
  
model.fit(X_train, y_train)
```

Prediction

```
y_pred = model.predict(X_test)
```

MODEL EVALUATION

Evaluation Metrics

The model was evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Evaluation Analysis

Write your actual interpretation based on your scores.

CONCLUSION

This project successfully demonstrated the application of machine learning to student performance prediction.

The complete machine learning workflow was implemented, including data collection, preprocessing, exploratory data analysis, feature selection, model training, prediction, and evaluation.

A Random Forest Classifier was developed to classify students into Pass and Fail categories. The model's performance was evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

The project demonstrates that machine learning can be used as a supporting tool for analyzing student performance and potentially identifying students who may require additional academic assistance.

FUTURE SCOPE

The project can be improved in several ways:

1. Larger Dataset

More student records can be collected to improve model generalization.

2. Additional Features

Attendance, assignment scores, online learning activity, and examination history could be included.

3. Algorithm Comparison

Random Forest could be compared with Logistic Regression, KNN, Decision Tree, and other algorithms.

4. Hyperparameter Optimization

Techniques such as Grid Search or Random Search could be used to improve model performance.

5. Web Application

The trained model could be deployed as a web application where teachers can enter student information and obtain a prediction.

6. Early Warning System

The system could be integrated into educational platforms to identify students who may need academic support at an early stage.

Output

```
print("=" * 50)
print("STUDENT PERFORMANCE PREDICTION")
print("=" * 50)

print(f"Accuracy   : {accuracy:.2%}")
print(f"Precision  : {precision:.2%}")
print(f"Recall     : {recall:.2%}")

print("\nModel: Random Forest Classifier")
print("Problem Type: Binary Classification")
print("Target: Pass / Fail")
print("=" * 50)
```

Output

STUDENT PERFORMANCE PREDICTION

Accuracy : XX.XX%

Precision : XX.XX%

Recall : XX.XX%

Model: Random Forest Classifier

Problem Type: Binary Classification

Target: Pass / Fail